

Domain-Adaptive Selective State-Space Learning for Cross-Subject Motor-Imagery EEG Decoding

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Abstract

Cross-subject motor-imagery electroencephalography (MI-EEG) decoding is limited by low signal-to-noise ratios, scarce labeled data, and strong distribution shifts across subjects and sessions. We propose a domain-adaptive selective state-space framework for this setting. A multi-kernel temporal convolutional front-end with depthwise spatial filtering and channel-group attention extracts local spectral-spatial features. Five selective state-space blocks with alternating forward and backward scans then model long-range temporal context at a cost linear in sequence length, and a convolutional shortcut with a grouped dilated temporal convolutional network fuses local and global information. To reduce cross-subject shift, the framework combines per-subject Riemannian alignment, gradient-reversal adversarial learning, and multi-kernel maximum mean discrepancy at the temporal and trial levels. At inference time, an information-maximization stage adapts only normalization parameters using unlabeled evaluation-session trials; their labels remain hidden until scoring. On BCI IV-2a, Ours achieved $75.81 \pm 10.63\%$ session-2 accuracy and a Cohen's κ score of 67.70 ± 14.20 on the $\times 100$ scale across nine LOSO target subjects. Code will be released at <https://github.com/USERNAME/REPOSITORY>.

Keywords: motor imagery, electroencephalography, brain-computer interface, selective state-space model, Mamba, domain adaptation, test-time adaptation

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1. Introduction

Motor-imagery brain-computer interfaces (MI-BCIs) decode movement intentions from non-invasive electroencephalography (EEG), with applications in rehabilitation, assistive communication, and human-machine control. Their principal signatures, event-related desynchronization and synchronization, are weak, non-stationary, and distributed across time and electrodes [1]. They also vary substantially among subjects and recording sessions. Consequently, a model trained on existing users often degrades on an unseen user, whereas acquiring labeled calibration trials for every user is time-consuming. Cross-subject decoding must therefore learn discriminative spatial-temporal structure from limited EEG while controlling inter-subject distribution shift without target labels.

Classical MI pipelines rely on common spatial pattern (CSP), which maximizes class-specific variance, and filter-bank CSP (FBCSP), which extends this idea across frequency bands [2, 3]. Although effective and interpretable, these methods separate filtering, feature extraction, and classification into manually designed stages; conventional CSP is also naturally binary and sensitive to the available trials and frequency range. Their subject-specific covariance structure can therefore transfer poorly when the user changes.

End-to-end convolutional neural networks reduce this dependence on handcrafted processing. Deep and shallow ConvNets learn temporal and electrode filters directly from EEG, while EEGNet uses depthwise and separable convolutions to obtain compact frequency-specific spatial features [4, 5]. Multi-scale temporal kernels are especially relevant because MI information is not confined to one frequency or duration. However, convolutions connect distant time points only through additional depth, larger kernels, or dilation, which may be insufficient to characterize the full temporal evolution of a trial.

Hybrid decoders therefore combine EEG-specific convolution with global sequence modeling. EEG Conformer applies self-attention after convolutional tokenization, ATCNet combines attention with a dilated temporal convolutional network (TCN), and TCFormer integrates a multi-kernel front-end, grouped-query Transformer layers, and a grouped TCN head [6, 7, 8]. These models demonstrate the value of combining local spectral-spatial priors with longer temporal context. Nevertheless, self-attention explicitly constructs pairwise token interactions and has quadratic sequence complexity [9]. This cost becomes undesirable for long, high-resolution EEG, while highly flexi-

ble attention operators may also be data-hungry in the small-sample regimes common in BCI studies.

Architecture alone cannot remove subject shift. Target-free domain generalization learns invariant representations from source subjects [10, 11, 12], whereas unsupervised domain adaptation (UDA) may additionally exploit unlabeled target EEG. The distinction is important: observing target signals without labels is UDA rather than zero-shot generalization. Existing UDA methods reduce covariance mismatch through Riemannian or Euclidean alignment [13, 14], minimize feature discrepancy with MK-MMD [15], or promote domain confusion through gradient reversal [16, 17]. HADANet combines adversarial and MK-MMD alignment for MI-EEG [18], while prototype-based methods additionally exploit local class structure [19]. These approaches remain vulnerable to negative transfer: marginal alignment may mix classes, and aligning only pooled trial features can overlook temporal mismatch.

Residual shift may also be addressed at inference. Information-maximization methods encourage confident but globally diverse predictions without labels [20], and TENT restricts adaptation to normalization parameters through entropy minimization [21]. Such constraints are attractive for EEG because target datasets are small and unrestricted fine-tuning can overfit or collapse to one predicted class. However, evaluation-session access must be disclosed explicitly, and labels must remain unavailable during adaptation and checkpoint selection.

Selective state-space models offer a linear-complexity alternative to attention. S4 introduced structured long-range sequence modeling [22], and Mamba made transition quantities input-dependent so that relevant information can be retained selectively [23]. Recent MI decoders have combined convolution and Mamba [24, 25], but primarily study architecture rather than leakage-controlled integration with target adaptation. Moreover, a generic state-space stack should complement, rather than replace, explicit temporal and electrode filtering. A purely forward scan also imposes unnecessary directionality when complete offline MI trials are available.

We therefore propose a target-aware framework coupling an EEG-specific encoder with hierarchical adaptation. Multi-kernel temporal and depthwise spatial convolutions extract local features; five alternating forward/backward selective state-space blocks model longer context; and a convolutional shortcut plus grouped TCN fuses local and global information. During training, subject-wise covariance alignment, a gradient-reversal discriminator, and trial- and temporal-level MK-MMD reduce target shift. At inference, in-

formation maximization updates only BatchNorm affine parameters using unlabeled evaluation-session EEG. Target labels remain hidden during pre-processing, training, checkpoint selection, and adaptation.

The main contributions are:

- an EEG encoder combining multi-scale convolution, alternating selective state-space modeling, local-global fusion, and a grouped dilated TCN;
- hierarchical cross-subject adaptation through leakage-safe covariance normalization, adversarial learning, and trial- and temporal-level MK-MMD;
- constrained label-free IM-TTA that updates only normalization affine parameters while preserving prediction diversity; and
- a chronological LOSO protocol that separates source-only, UDA, and test-time adaptation settings without target-label leakage.

2. Datasets

Three public MI-EEG datasets are used to examine whether the same decoder transfers across different numbers of electrodes, classes, subjects, and recording sessions. Dataset access and event extraction are implemented through the MOABB/Braindecode interface; this choice provides a common representation of subjects, sessions, runs, and event windows and supports reproducible benchmarking [26]. Table 1 summarizes the properties relevant to the present work.

Table 1: Summary of the MI-EEG datasets (EOG channels excluded).

Dataset	Subjects	Channels	Sessions	Classes
BCI IV-2a	9	22	2	left hand, right hand, feet, tongue
BCI IV-2b	9	3	5	left hand, right hand
Zhou2016	4	14	3	left hand, right hand, feet

2.1. BCI Competition IV-2a

BCI Competition IV dataset 2a contains recordings from nine participants. Each participant completed two sessions on different days, and each session contains 288 trials distributed equally over imagined left-hand, right-hand, both-feet, and tongue movements. Signals were acquired from 22 EEG electrodes and three EOG electrodes at 250 Hz; only the EEG electrodes are supplied to the classifier. The original recordings were hardware band-pass filtered between 0.5 and 100 Hz and notch filtered at 50 Hz [27]. The implementation uses the event-defined MI interval exposed by MOABB, rescales volts to microvolts, and does not apply an additional frequency filter. The first recording session is denoted session 1 and the second is denoted session 2 throughout this paper.

2.2. BCI Competition IV-2b

BCI Competition IV dataset 2b contains two-class left- versus right-hand MI recordings from nine participants. Three bipolar EEG derivations centered on C3, Cz, and C4 were sampled at 250 Hz, together with three EOG channels that are excluded from decoding. Each participant completed five sessions: the first three form the calibration portion and the final two form the original evaluation portion [27]. The present loader follows this natural chronological separation. The event-defined MI window is used after removing the final 0.5 s through the configured stop offset. This dataset tests whether the method remains useful when spatial coverage is reduced from 22 electrodes to only three.

2.3. Zhou2016

The Zhou2016 dataset contains four participants, each recorded in three sessions separated by days to months. A session consists of two runs of 75 trials, with 25 trials per class, giving 450 trials per participant. The three MI classes are left hand, right hand, and feet. EEG was recorded from 14 electrodes at 250 Hz. Each trial contains a 5-s visual arrow period during which the indicated imagery is performed, followed by a 4-s rest period [28]. We therefore extract exactly the first 5.0 s from cue onset rather than including the subsequent rest interval. The public data are loaded through the MOABB representation, whose consistent dataset API is useful for avoiding dataset-specific indexing errors [26].

3. Proposed Method

Figure 1 summarizes one LOSO fold. A shared encoder processes labeled source trials and unlabeled target trials during Stage 1 UDA. After source-validation checkpoint selection, Stage 2 adapts only normalization parameters with unlabeled evaluation-session EEG; labels are revealed only for Stage 3 scoring.

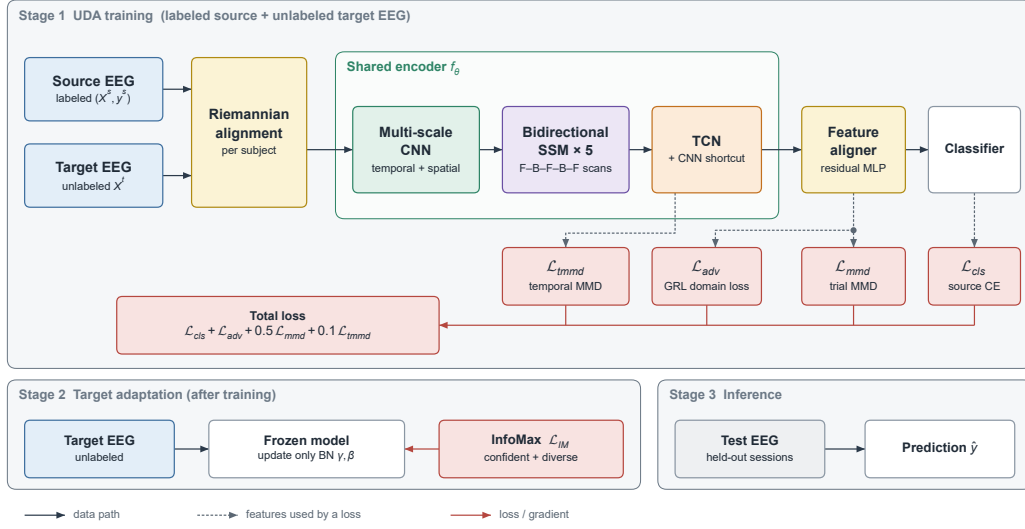


Figure 1: Framework for one LOSO fold: joint source classification and target adaptation (Stage 1), label-free IM-TTA (Stage 2), and scoring with previously withheld labels (Stage 3).

3.1. Problem formulation

Let $\mathcal{S} = \{(X_i^s, y_i^s)\}_{i=1}^{N_s}$ be labeled trials from the source subjects, $\mathcal{T}_a = \{X_j^a\}_{j=1}^{N_a}$ the earlier unlabeled target sessions used for UDA, and $\mathcal{T}_e = \{X_j^e\}_{j=1}^{N_e}$ the later target sessions used for evaluation, where $X \in \mathbb{R}^{C \times T}$. Training uses $\mathcal{S}$ and $\mathcal{T}_a$; IM-TTA may access only the signal values of $\mathcal{T}_e$, never their labels.

3.2. Leakage-safe signal alignment

A separate Riemannian whitening operator is optionally fitted for each subject. OAS trial covariances are summarized by their affine-invariant mean

$\bar{R}$, and trials are transformed as

$$\tilde{X} = \bar{R}^{-1/2} X. \quad (1)$$

A source transform is fitted on that subject’s training sessions and applied to its validation sessions; the target transform is fitted on $\mathcal{T}_a$ and frozen before application to $\mathcal{T}_e$ [13, 14]. Evaluation-session data therefore update neither alignment nor standardization statistics.

3.3. Multi-scale convolutional encoder

As detailed in Fig. 2, three temporal convolutions with kernels of 20, 32, and 64 samples produce 32 maps each [8]. Their concatenated output is processed by depthwise convolution across all electrodes with depth multiplier two, average pooling, and dropout. A grouped pointwise projection reduces the 192 spatial maps to 48 channels (16 per temporal scale), followed by grouped temporal convolution and residual channel-group attention. The latter assigns one learned coefficient to each scale. A second pooling layer yields $H_c \in \mathbb{R}^{B \times 48 \times L}$.

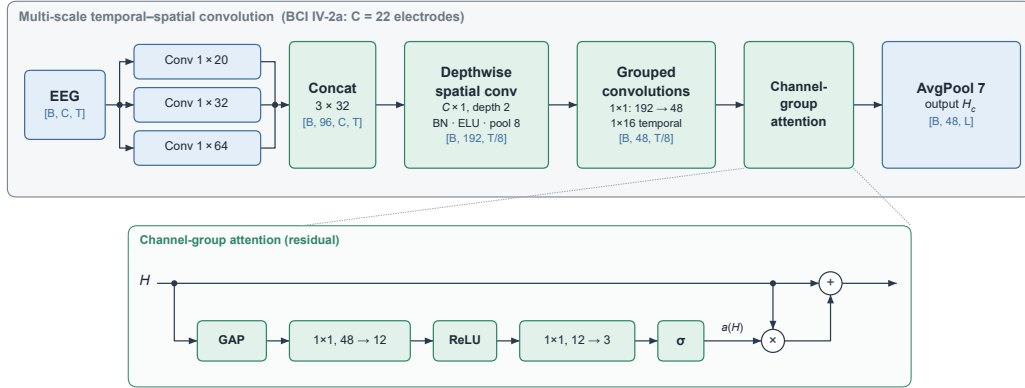


Figure 2: Multi-scale temporal-spatial front-end. Three temporal branches, depthwise electrode filtering, grouped refinement, and channel-group attention produce H_c .

3.4. Alternating selective state-space encoder

After pointwise channel mixing, five selective state-space blocks scan in the order forward-backward-forward-backward-forward. A backward block reverses the temporal axis before and after its recurrence, providing bidirectional offline context without duplicating every block. Within each block,

layer normalization, gated projections, and a causal depthwise convolution generate u_t . A diagonal selective recurrence then computes [22, 23]

$$A = -\exp(A_{\log}), \quad (2)$$

$$h_t = \exp(\Delta_t A) \odot h_{t-1} + \Delta_t B_t \odot u_t, \quad (3)$$

$$z_t = C_t^\top h_t + d_{\text{skip}} \odot u_t. \quad (4)$$

Here Δ_t , B_t , and C_t depend on the input. The implementation uses state dimension eight, dropout 0.4, and depth-dependent stochastic depth. It is a lightweight PyTorch selective SSM rather than the official hardware-aware Mamba kernel, but retains linear scaling in sequence length.

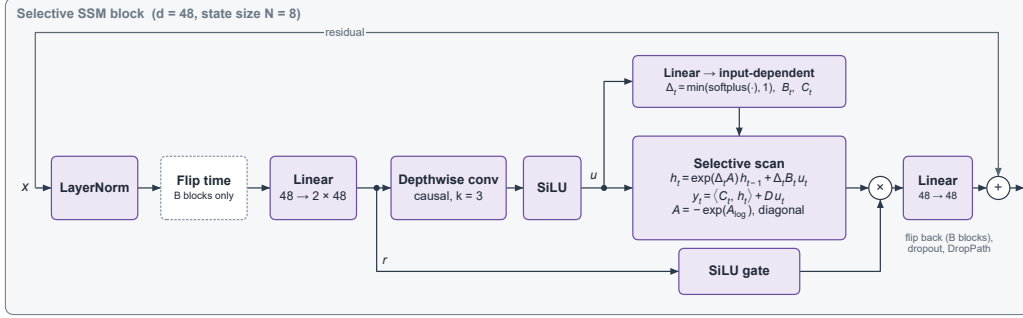


Figure 3: Selective SSM block. Backward blocks reverse and restore the temporal axis around the mixer; forward blocks omit these operations.

3.5. Local-global fusion and classification

The state-space output is reduced from 48 to 16 channels and concatenated with the 48-channel shortcut H_c . Two residual grouped TCN blocks with kernel length four and dilations one and two refine the resulting 64-channel sequence. The mean of its last state and temporal average forms the trial representation. A norm-constrained grouped 1×1 classifier produces four group-wise logits, which are combined by learned softmax weights (Fig. 4).

3.6. Hierarchical unsupervised domain adaptation

A residual $64 \rightarrow 128 \rightarrow 64$ feature aligner precedes classification and domain alignment. Source cross-entropy $\mathcal{L}_{\text{cls}}$ maintains discrimination. A

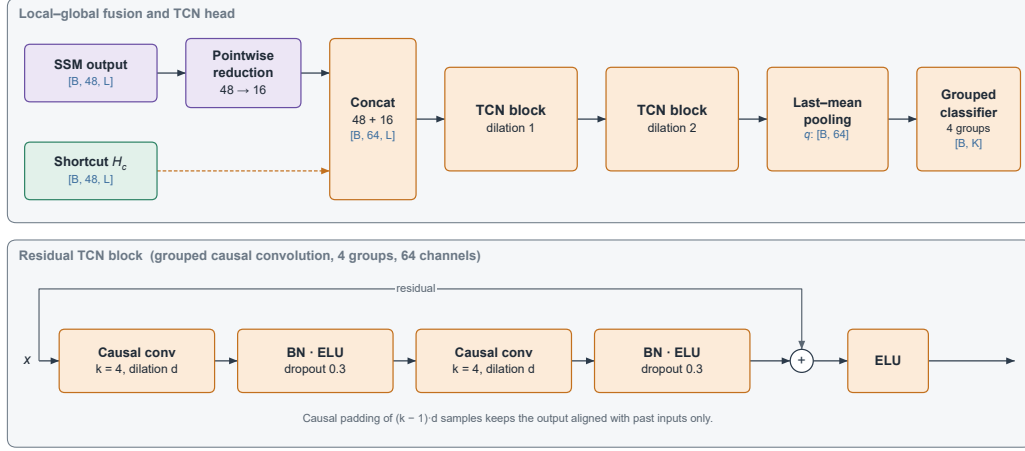


Figure 4: Local-global fusion, two-block grouped TCN, temporal aggregation, and grouped classification.

gradient-reversal discriminator provides domain loss $\mathcal{L}_{\text{adv}}$ [16], while five-kernel MK-MMD aligns normalized source and target representations [15]. It is applied both to pooled trial features ($\mathcal{L}_{\text{mmd}}$) and to time-averaged TCN sequences ($\mathcal{L}_{\text{tmmd}}$), giving

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{adv}} + 0.5\mathcal{L}_{\text{mmd}} + 0.1\mathcal{L}_{\text{tmmd}}. \quad (5)$$

Only $\mathcal{L}_{\text{cls}}$ uses labels, all of which come from source subjects. The discriminator coefficient increases smoothly from zero to one with training progress.

3.7. Constrained information-maximization adaptation

After checkpoint selection, all parameters except BatchNorm affine weights and biases are frozen. Five Adam passes over unlabeled $\mathcal{T}_e$ at learning rate 10^{-4} minimize

$$\mathcal{L}_{\text{IM}} = \frac{1}{N_e} \sum_j H(p_j) - H\left(\frac{1}{N_e} \sum_j p_j\right), \quad (6)$$

where the first term encourages confident predictions and the second preserves class diversity [20, 21]. No evaluation label or metric is used for adaptation or checkpoint selection.

4. Experimental Protocol

4.1. Chronological LOSO protocol with test-time adaptation

One subject is held out in each LOSO fold. All remaining subjects are sources; the held-out participant is the target. Table 2 specifies the chronological session split. Earlier target sessions are used without labels during Stage 1 UDA. The later evaluation sessions remain unavailable during model training, preprocessing-statistic fitting, and checkpoint selection; at inference time, their EEG signals are passed without labels to IM-TTA, after which the adapted model is scored using the withheld labels. This is the primary protocol for headline comparisons.

Table 2: Chronological session allocation in each LOSO fold. Evaluation-session EEG is available without labels only to IM-TTA at inference time; its labels are used afterward solely for scoring.

Dataset	Source subjects (labeled)		Target subject	
	Training	Validation	Stage 1 UDA (unlabeled)	IM-TTA + evaluation (labels withheld)
BCI IV-2a	session 1	session 2	session 1	session 2
BCI IV-2b	sessions 1–3	sessions 4–5	sessions 1–3	sessions 4–5
Zhou2016	sessions 1–2	session 3	session 1	sessions 2–3

For BCI IV-2a, this split yields nine folds; for BCI IV-2b, nine folds; and for Zhou2016, four folds. The target identifier changes between folds, but the split rule and hyperparameters remain fixed. The checkpoint with the highest source-validation accuracy is selected before IM-TTA. Neither evaluation-session labels nor evaluation metrics participate in adaptation or checkpoint selection. Stage 1 adversarial/MK-MMD training uses the earlier unlabeled target-adaptation loader, whereas Stage 2 IM-TTA uses the later unlabeled evaluation loader immediately before predictions are scored.

4.2. Preprocessing and augmentation

Only EEG channels are retained. Signals are resampled to 250 Hz and converted from volts to microvolts. No additional digital band-pass filter is applied in the model configuration, because the public BCI Competition

recordings already include acquisition filtering and the network learns its temporal filters from the signal. BCI IV-2a uses the complete MOABB event window; BCI IV-2b removes 0.5 s from the end of the event window; Zhou2016 uses exactly 5.0 s beginning at cue onset.

Riemannian whitening, when active, follows the session-isolated fitting rule in Section 3.2. Z-score statistics are fitted using the source-training partition and then frozen. Inter-trial segment reconstruction augmentation is applied only to labeled source mini-batches: segments are recombined among trials of the same source class, so it neither reads target labels nor creates mixed source–target examples. Zhou2016 is run without this augmentation to preserve its intended dataset-specific setting.

4.3. Training configuration

All experiments are executed in Kaggle notebooks using one NVIDIA Tesla T4 GPU with 16 GB memory. BCI IV-2a and BCI IV-2b use a mini-batch size of 48 and a maximum of 125 epochs. Zhou2016 uses a mini-batch size of 32 and 100 epochs. The BCI Competition runs use Adam with an initial learning rate of 9×10^{-4} , $\beta_1 = 0.5$, and weight decay 10^{-3} ; the Zhou2016 setting uses an initial learning rate of 10^{-3} and weight decay 10^{-4} . A linear warm-up over the first three LOSO epochs is followed by cosine decay. The random seed, data split, and initialization procedure are held fixed for direct architectural ablations. Final reporting should additionally repeat each complete LOSO experiment with multiple seeds to quantify optimization variance.

4.4. Evaluation measures and reporting

The primary measures are trial-level classification accuracy and Cohen’s κ , reported as mean and standard deviation across target-subject folds; all reported κ values are multiplied by 100. Per-subject accuracies are reported for every dataset so that a mean is not interpreted without its cross-subject dispersion. Accuracy differences between the proposed method and each baseline are tested on paired subject folds with a two-sided Wilcoxon signed-rank test. Parameter count, single-trial inference latency, and throughput are secondary efficiency measures. In the primary protocol, evaluation-session signals may be used without labels only by the explicitly identified IM-TTA stage; results over the union of earlier adaptation sessions and later evaluation sessions are marked auxiliary.

Inference efficiency is measured after training on the same NVIDIA Tesla T4 GPU used for training, with preprocessing and host-device transfer excluded. Every network is placed in evaluation mode and executed under inference-only semantics with the BCI IV-2a input shape (22 channels $\times$ 1000 samples). Single-trial latency is measured with batch size one and explicit CUDA synchronization; each measurement uses 100 warm-up iterations followed by 1000 timed iterations. Throughput is measured independently with a common batch size of 32 and is reported in trials per second; it is not inferred by taking the reciprocal of batch-one latency. Information-maximization adaptation is a one-time target-subject operation and is therefore timed separately in seconds per target subject rather than included in inference latency.

4.5. Baseline comparison policy

Tables 3–5 group target-free source-only methods separately from methods that use unlabeled target EEG. Every baseline must use the same subject folds, chronological session allocation, and target-data access specified in Table 2. Stage 1 UDA methods may use the designated earlier target sessions without labels; methods evaluated with test-time adaptation may additionally use the later evaluation-session signals without labels immediately before prediction. Evaluation labels remain unavailable for training, adaptation, preprocessing, checkpoint selection, and hyperparameter selection. Published headline values obtained under incompatible splits or target-label access are not copied into the tables.

The source-only group comprises three deep decoders that combine convolution with attention or temporal convolution: EEG Conformer [6], ATCNet [7], and TCFormer [8]. These models receive no target EEG before final inference. The UDA group includes an MK-MMD/DAN baseline [15], and conditional adversarial adaptation [17].

The proposed method is reported in three variants that share the same encoder and differ only in how much target EEG they use (Fig. 1):

- **Ours-S** (source-only): trained with $\mathcal{L}_{\text{cls}}$ on the labeled source subjects only. No target EEG is seen before inference, so this variant is directly comparable with the source-only baselines and isolates the contribution of the encoder.
- **Ours-U** (UDA): Stage 1 only, trained with the complete objective in Eq. (5) using the unlabeled target-adaptation EEG, without

information-maximization adaptation.

- **Ours** (complete framework): Ours-U followed at inference time by Stage 2 information-maximization adaptation of BatchNorm affine parameters on the unlabeled evaluation-session EEG (Section 3.7); labels are revealed only for scoring after adaptation.

The difference between Ours-U and Ours-S therefore measures the gain from domain alignment, and the difference between Ours and Ours-U measures the gain from information-maximization adaptation.

5. Results

Table 3 contains the completed BCI IV-2a result for Ours; entries requiring rerun baselines, repeated seeds, or statistical tests remain unpopulated. Results are reported per target subject together with the mean and standard deviation across the nine LOSO folds. The standard deviations therefore quantify between-subject dispersion for this run, not variability across random seeds.

5.1. BCI Competition IV-2a

Table 3 reports the four-class chronological-session LOSO results for the nine BCI IV-2a target subjects. Ours applies label-free IM-TTA to each target subject’s session 2 EEG immediately before evaluation.

For this completed nine-fold run, Ours achieved a mean session-2 accuracy of $75.81 \pm 10.63\%$, a mean Cohen’s κ score of 67.70 ± 14.20 on the $\times 100$ scale, and a mean cross-entropy loss of 0.797 ± 0.404 . The model contains 131,890 trainable parameters, and aggregate training across all nine folds required 572.23 minutes on a Kaggle NVIDIA Tesla T4. The pipeline-reported response time was 28.98 ms per trial; this value is not entered in Table 6 because the current timing routine explicitly moves the model to the CPU, whereas that table requires a controlled T4 measurement. Under the auxiliary sessions 1+2 view, Ours obtained $76.10 \pm 9.77\%$ accuracy, a Cohen’s κ score of 68.10 ± 13.00 on the $\times 100$ scale, and loss 0.814 ± 0.416 .

5.2. BCI Competition IV-2b

Table 4 reports the binary left- versus right-hand results for the nine BCI IV-2b target subjects, whose recordings contain only three bipolar EEG derivations.

Table 3: Per-subject LOSO accuracy (%) and Cohen’s κ multiplied by 100 on BCI IV-2a (four classes). Ours-S: source-only; Ours-U: UDA without IM adaptation; Ours: UDA with IM adaptation (Section 4.5). p : Wilcoxon signed-rank test of accuracy against Ours.

Subj.		Source-only				UDA			Ours
		Conformer	ATCNet	TCFormer	Ours-S	DAN	CDAN	Ours-U	
		[6]	[7]	[8]		[15]	[17]		
S1	Acc.	—	—	—	—	—	—	—	78.82
	κ	—	—	—	—	—	—	—	71.76
S2	Acc.	—	—	—	—	—	—	—	57.64
	κ	—	—	—	—	—	—	—	43.52
S3	Acc.	—	—	—	—	—	—	—	93.75
	κ	—	—	—	—	—	—	—	91.67
S4	Acc.	—	—	—	—	—	—	—	74.31
	κ	—	—	—	—	—	—	—	65.74
S5	Acc.	—	—	—	—	—	—	—	65.62
	κ	—	—	—	—	—	—	—	54.17
S6	Acc.	—	—	—	—	—	—	—	66.67
	κ	—	—	—	—	—	—	—	55.56
S7	Acc.	—	—	—	—	—	—	—	85.42
	κ	—	—	—	—	—	—	—	80.56
S8	Acc.	—	—	—	—	—	—	—	84.38
	κ	—	—	—	—	—	—	—	79.17
S9	Acc.	—	—	—	—	—	—	—	75.69
	κ	—	—	—	—	—	—	—	67.59
Mean	Acc.	—	—	—	—	—	—	—	75.81
	κ	—	—	—	—	—	—	—	67.70
SD	Acc.	—	—	—	—	—	—	—	10.63
	κ	—	—	—	—	—	—	—	14.20
p		—	—	—	—	—	—	—	—

Table 4: Per-subject LOSO accuracy (%) and Cohen’s κ multiplied by 100 on BCI IV-2b (two classes). Ours-S: source-only; Ours-U: UDA without IM adaptation; Ours: UDA with IM adaptation (Section 4.5). p : Wilcoxon signed-rank test of accuracy against Ours.

Subj.		Source-only				UDA			
		Conformer	ATCNet	TCFormer	Ours-S	DAN	CDAN	Ours-U	Ours
		[6]	[7]	[8]		[15]	[17]		
S1	Acc.	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—
S2	Acc.	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—
S3	Acc.	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—
S4	Acc.	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—
S5	Acc.	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—
S6	Acc.	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—
S7	Acc.	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—
S8	Acc.	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—
S9	Acc.	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—
Mean	Acc.	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—
SD	Acc.	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—
p		—	—	—	—	—	—	—	—

5.3. Zhou2016

Table 5 reports the currently available source-only and no-IM-TTA results for the four Zhou2016 target subjects. Both populated variants are evaluated over all three target sessions and are therefore auxiliary all-session results rather than primary chronological comparisons. With only four folds, the smallest attainable two-sided Wilcoxon p -value is 0.125.

Table 5: Per-subject LOSO accuracy (%) and Cohen’s κ multiplied by 100 on Zhou2016 (three classes). Ours-S: source-only; Ours-U: UDA without IM adaptation; Ours: UDA with IM adaptation. The populated Ours-S and Ours-U values use the auxiliary all-three-session target view.

Subj.		Source-only				UDA			
		Conformer	ATCNet	TCFormer	Ours-S	DAN	CDAN	Ours-U	Ours
		[6]	[7]	[8]		[15]	[17]		
S1	Acc.	—	—	—	73.90	—	—	72.86	—
	κ	—	—	—	60.86	—	—	59.29	—
S2	Acc.	—	—	—	77.93	—	—	81.61	—
	κ	—	—	—	66.90	—	—	72.41	—
S3	Acc.	—	—	—	62.97	—	—	67.18	—
	κ	—	—	—	44.41	—	—	50.77	—
S4	Acc.	—	—	—	76.32	—	—	76.09	—
	κ	—	—	—	64.48	—	—	64.14	—
Mean	Acc.	—	—	—	72.78	—	—	74.44	—
	κ	—	—	—	59.20	—	—	61.70	—
SD	Acc.	—	—	—	5.84	—	—	5.23	—
	κ	—	—	—	8.80	—	—	7.80	—
p		—	—	—	—	—	—	—	—

For the four-fold Ours-S run, mean all-session accuracy was $72.78 \pm 5.84\%$, mean Cohen’s κ was 59.20 ± 8.80 on the $\times 100$ scale, and mean loss was 0.770 ± 0.293 . The source-only model contained 96,812 trainable parameters. Training all folds required 48.38 minutes on a Kaggle NVIDIA Tesla T4, with a mean all-session test time of 1.01 s per subject and a pipeline-reported response time of 27.31 ms per trial. For the corresponding Ours-U run, mean all-session accuracy was $74.44 \pm 5.23\%$, mean Cohen’s κ was 61.70 ± 7.80 , and mean loss was 0.763 ± 0.194 . Ours-U contained 130,286 trainable parameters

and required 70.61 minutes for all folds, with a mean test time of 0.99 s per subject and a response time of 26.88 ms per trial. These pipeline-reported response times are excluded from Table 6 because they were not obtained under that table’s controlled T4 timing protocol.

5.4. Computational efficiency

Table 6 compares model size and inference cost on the Tesla T4 GPU.

Table 6: Computational efficiency on a Tesla T4 GPU (BCI IV-2a input, 22×1000). Ours-S, Ours-U, and Ours are defined in Section 4.5.

Method	Parameters (M)	Latency (ms/trial)	Throughput (trials/s)	IM adaptation (s/subject)
EEG Conformer [6]	—	—	—	N/A
ATCNet [7]	—	—	—	N/A
TCFormer [8]	—	—	—	N/A
Ours-S	—	—	—	N/A
Ours-U	—	—	—	N/A
Ours	0.132	—	—	—

6. Ablation Study

Table 7 uses the auxiliary all-session target view (sessions 1–2 for BCI IV-2a, 1–5 for BCI IV-2b, and 1–3 for Zhou2016) and therefore remains separate from the primary chronological evaluation. Each run changes one component while preserving the folds, seeds, training budget, checkpoint rule, and label-free IM-TTA protocol. The variants remove Riemannian alignment, channel-group attention, alternating scans, the grouped TCN, adversarial/MMD alignment, temporal MMD, or IM-TTA, thereby separating architectural from adaptation gains.

Table 7: Ablation results under the auxiliary all-session protocol: mean accuracy (%) and Cohen’s κ multiplied by 100; available Ours values are mean $\pm$ standard deviation across target subjects.

Configuration	BCI IV-2a		BCI IV-2b		Zhou2016	
	Acc.	κ	Acc.	κ	Acc.	κ
Ours (complete framework)	76.10 $\pm$ 9.77	68.10 $\pm$ 13.00	–	–	–	–
w/o Riemannian alignment	–	–	–	–	–	–
w/o channel-group attention	–	–	–	–	–	–
Forward-only state-space scans	–	–	–	–	–	–
w/o grouped TCN	–	–	–	–	–	–
w/o adversarial and MMD losses	–	–	–	–	–	–
w/o temporal MMD	–	–	–	–	–	–
w/o information-maximization	–	–	–	–	74.44 $\pm$ 5.23	61.70 $\pm$ 7.80

7. Discussion

On BCI IV-2a, Ours achieved $75.81 \pm 10.63\%$ session-2 accuracy and 67.70 ± 14.20 Cohen’s κ on the $\times 100$ scale. The large between-subject dispersion, including accuracies from 57.64% to 93.75%, shows that the mean does not describe every target equally well and motivates subject-wise reporting. The auxiliary sessions 1+2 result ($76.10 \pm 9.77\%$) is not directly comparable because it also scores the earlier session exposed without labels during Stage 1 UDA. For Zhou2016, Ours-S reached $72.78 \pm 5.84\%$ accuracy and 59.20 ± 8.80 Cohen’s κ over all three sessions, whereas the no-IM-TTA Ours-U variant reached $74.44 \pm 5.23\%$ and 61.70 ± 7.80 , respectively. The resulting gains of 1.66 accuracy points and 2.50 κ points quantify the contribution of Stage 1 UDA in this auxiliary all-session view; the contribution of IM-TTA remains unestablished until the complete Ours variant is evaluated.

The architecture combines complementary biases: multi-scale and depth-wise convolutions extract local spectral–spatial patterns, alternating selective scans provide linear-cost offline context, and the shortcut/TCN retains local evidence. Likewise, adversarial learning and MK-MMD provide learned and explicit distribution alignment, while the lower-weight temporal term regularizes sequence features. Their individual contributions must be established by the ablations rather than inferred from the complete result. Alternating scans are appropriate for offline trials but are non-causal; online deployment would require forward-only scans and hardware-specific latency evaluation.

The method remains transductive because IM-TTA accesses unlabeled

evaluation-session signals before scoring, although labels remain hidden. It must therefore be compared only with methods receiving the same target access. Other limitations are the three-channel setting of BCI IV-2b, the four subjects in Zhou2016, dataset-specific windows and training budgets, and the currently incomplete repeated-seed and matched-baseline results. Final claims should be based on completed ablations, multiple seeds, paired tests, and adaptation-time measurements across all three datasets.

8. Conclusion

We presented a cross-subject MI-EEG framework combining multi-scale spatial-temporal convolution, alternating selective state-space blocks, local-global TCN fusion, hierarchical UDA, and constrained IM-TTA. Its chronological LOSO protocol excludes target labels from preprocessing, training, adaptation, and checkpoint selection. On BCI IV-2a, Ours obtained $75.81 \pm 10.63\%$ session-2 accuracy and 67.70 ± 14.20 Cohen’s κ on the $\times 100$ scale. Complete-framework results on Zhou2016 and BCI IV-2b, matched baselines, repeated seeds, and ablations are required before drawing broader conclusions.

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